# Nexa Drive: An On-Device Vietnamese In-Car Assistant with Distilled Small Language Models and Functional Tokens

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## Abstract

We present **Nexa Drive**, a Vietnamese voice assistant that controls 24 functions of a simulated car and runs entirely on a laptop CPU. Following Octopus v2, every function is a dedicated vocabulary token, so choosing a function is a single-token decision and the prompt shrinks from about 2,150 tokens to 33. A Qwen2.5-1.5B-Instruct student is trained with LoRA on 33.5k examples distilled from a locally hosted Qwen2.5-72B-Instruct teacher and filtered by code checks, a two-family LLM jury and human audits. Besides executing or refusing, the model can *clarify*: an implicit complaint such as *"nóng quá"* (*it is so hot*) becomes a proposal the driver confirms. Across six data rounds evaluated on frozen, contamination-checked test sets with two seeds, exact McNemar tests and criteria fixed before each run, we find that (i) adding a sentence style to only one label creates a shortcut that contrastive pairs remove; (ii) the student is statistically indistinguishable from the prompted 72B teacher on clear, short, misspelled and long implicit requests, but refuses out-of-scope requests less reliably (77.4% vs. 94.9%) and resolves two-turn follow-ups worse (62.5% vs. 81.7%); (iii) a confidence threshold on the decision token cuts wrong actions on the car by 29–40%; (iv) GRPO on saturated distilled data does not reduce wrong actions; and (v) 4-bit quantization passes a validation check yet costs about five points on implicit requests at test time. With real speech from one speaker the full pipeline is correct on 63.3% of new commands (70.0% when typed), tolerates recorded in-car noise down to 0 dB, and is woken by a custom "Hey Nexa" detector that catches 39 of 40 real calls at 2.86 false alarms per hour. Fine-tuning the recognizer with synthetic commands plus 12 minutes of that speaker's voice raises correct commands on a further new set from 24 to 28 of 40, whereas a faster recognizer with lower word error rate, confidence-based fusion and keyword repair do not help.

## 1 Introduction

A driver should be able to say *"bật điều hoà 22 độ"* (*turn the AC on at 22 degrees*) or simply *"trong xe bí quá"* (*it is stuffy in here*) and have the car respond, without taking their hands off the wheel and without depending on a mobile connection. Large language models (LLMs) can map such utterances to structured function calls, but the usual recipe—placing the schemas of every available function in the prompt of a large model—is slow and memory-hungry on the kind of hardware found in a car, and requires a network connection when the model is hosted remotely.

Octopus v2 (Chen and Li, 2024) proposed a different recipe for on-device agents: each function becomes a new *functional token*, a small model is fine-tuned to emit `<nexa_i>(arguments)`, and the function descriptions are only needed at training time. Function selection then reduces to predicting one token, and the inference prompt contains little more than the user's sentence. Follow-up work applied small models to in-vehicle function calling (Sowti Khiabani et al., 2025; Jafarzadeh Asl et al., 2026) and showed a clear trade-off: functional tokens are many times faster than schema-in-prompt, but tend to call a function when they should decline (Jafarzadeh Asl et al., 2026).

This report describes **Nexa Drive**, a graduation project that builds such an assistant for *Vietnamese* and takes it all the way to a working product: speech recognition, a wake word, runtime guardrails, speech synthesis and an interactive 3D car. The product is the goal; the research layer exists to justify each technical choice with measurements. The setting raises four problems that published function-calling work rarely addresses together:

1. **No Vietnamese in-car data.** We generate all training data with an open 72B teacher run locally on rented GPUs (no paid API), and eval-

uate on separately generated, human-reviewed sets that are frozen by hash before any model is scored.

2. **Drivers rarely speak in commands.** Many requests are complaints (*"nóng quá"* (*it is so hot*)) whose right answer is neither a call nor a refusal but a proposal. We add a third action, *clarify*, marked by a `<nexa_ask>` token, and measure it on dedicated sets.
3. **A wrong action moves a physical car.** We keep the model responsible for understanding and put safety in code: a confidence threshold on the decision token, a parameter validator, state-dependent rules (no opening the trunk while driving) and confirmation for sensitive actions or wake-word commands.
4. **Speech is the main interface.** Errors of the speech recognizer, Vietnamese number words, background noise and false wake-ups all reach the language model, so we evaluate the whole chain with real recordings.

**Contributions.**

- A complete on-device Vietnamese in-car assistant: a 1.5B functional-token model (1.04 GB after 4-bit quantization, median 0.46 s per decision on a laptop CPU), speech recognition, speech synthesis, a custom wake word, runtime guardrails and a 3D interface, all running offline (§3, §7).
- A distillation pipeline with code-first validation, a two-family LLM jury and human audits, together with a record of what did and did not work over six data rounds—most notably that hard examples must come in *contrastive pairs* across labels, or the model learns a shortcut from sentence form (§4, §6.2).
- An evaluation protocol for small test sets: frozen and contamination-checked sets, two training seeds per configuration, exact McNemar tests and acceptance criteria written down before each run. Under this protocol several apparent improvements turned out to be seed noise, and we report them as such (§5).
- Measured answers to practical questions: how a 1.5B student compares with its 72B teacher (§6.1), LoRA versus full fine-tuning (§6.3), what a confidence threshold buys (§6.5), what 4-bit quantization costs (§6.6), whether GRPO helps on distilled data (§6.7), and how much speech recognition and noise degrade the chain (§7). Negative results are reported alongside positive ones.

The vehicle is simulated: the assistant drives a state machine and a 3D model rather than a real CAN bus. All evaluation sets are synthetic (generated by an LLM from a written specification and reviewed by a human), and all voice recordings come from one speaker. We return to these limitations at the end of the report.

## 2 Related Work

**Function calling and tool use.** LLMs learn to call tools from self-generated demonstrations (Schick et al., 2023), from API documentation (Patil et al., 2024) or from large synthetic corpora over thousands of APIs (Qin et al., 2024). APIGen (Liu et al., 2024b) and ToolACE (Liu et al., 2024a) show that synthetic function-calling data needs layered verification—format checks, execution checks and model-based semantic checks—and that samples failing a check should be discarded. BFCL (Patil et al., 2025) is the de facto benchmark and measures, among others, irrelevance detection. These works place the available functions in the prompt; our model instead memorizes a fixed set of functions as tokens. When2Call (Ross et al., 2025) studies when a model should call, ask a follow-up question or decline, and finds open models weak at this decision. Our clarify action targets a related but different case: the user does not issue a command at all but describes a situation.

**Small and on-device agents.** Octopus v2 (Chen and Li, 2024) introduced functional tokens and reported 99.5% accuracy for a 2B model on Android functions, evaluated on queries produced by the same generator as the training data. TinyAgent (Erdogan et al., 2024) runs a 1.1B model on a laptop with 4-bit quantization and a tool retriever. Hammer (Lin et al., 2024b) targets on-device function calling and improves irrelevance detection with dedicated data. For cars, Sowti Khiabani et al. (2025) deploy a quantized Phi-3 model with functional tokens on a vehicle head unit, and Jafarzadeh Asl et al. (2026) compare functional tokens with schema-in-prompt on 79 functions derived from Android Automotive: accuracy on known functions is similar, functional tokens are 26–98× faster, but over-trigger on 17.4% of out-of-scope queries versus 1.6–4.3%. CAR-bench (Kirmayr et al., 2026) and VehicleWorld (Yang et al., 2025) provide English in-car agent environments

with tools in the prompt.

**Distillation and data quality.** We use sequence-level knowledge distillation (Hinton et al., 2015; Kim and Rush, 2016): the student is trained on outputs of a larger teacher. To judge generated data we follow reference-guided judging (Zheng et al., 2023), a jury of models from different families (Verga et al., 2024) and agreement measured with Cohen's $\kappa$ rather than raw agreement (Thakur et al., 2025). For diversity we vary personas (Ge et al., 2024) and situational context (Xu et al., 2024) at each generation call and reject near-duplicates (Wang et al., 2023).

**Abstention and confidence.** Thresholding the maximum softmax probability is a standard baseline for detecting errors and out-of-distribution inputs (Hendrycks and Gimpel, 2017), and selective classification trades coverage for risk (Geifman and El-Yaniv, 2017); intent classifiers face the same out-of-scope problem (Larson et al., 2019). Conformal methods give distribution-free guarantees when calibration data match deployment (Angelopoulos and Bates, 2021). Function-calling papers mostly address over-triggering with data; with functional tokens the decision is a single token, so its probability is directly available.

**Reinforcement learning for tool use.** GRPO (Shao et al., 2024) optimizes a policy with group-normalized, rule-based rewards; DAPO (Yu et al., 2025) filters groups without reward variance during training. ToolRL (Qian et al., 2025) and Tool-N1 (Zhang et al., 2025) apply GRPO to tool calling with an explicit reasoning step before the call. LoRA (Hu et al., 2022) appears sufficient for RL fine-tuning (Schulman and Thinking Machines Lab, 2025).

**Guardrails and in-car safety.** NeMo Guardrails (Rebedea et al., 2023) enforces human-written rules outside the model. $\tau$-bench (Yao et al., 2024) requires agents to obtain explicit confirmation before state-changing actions, and ToolSandbox (Lu et al., 2024) tests whether agents ask for missing information instead of guessing. For cars, Afroze et al. (2026) find that LLM authorization decisions are not reliable enough to be the only safety mechanism and recommend an independent enforcement layer; Bouchekir et al. (2026) compile natural-language policies into executable checks run before each tool call; DriveSafe (Kumar et al., 2026) provides a risk taxonomy for driving assistants. Voice interfaces can also be attacked with commands the driver does not hear (Zhang et al., 2017).

**Spoken function calling.** SLURP (Bastianelli et al., 2020) benchmarks spoken language understanding for home assistants. SFC-Bench (Peng et al., 2026) measures function calling from speech: on its easiest level, Qwen3-32B drops from 81.7% with text to 64.6% with transcribed speech, and the authors identify misrecognized names and numbers as the main bottleneck. VoiceAgentBench (Jain et al., 2025) finds cascaded ASR+LLM pipelines ahead of open speech LLMs, with larger drops for Indic languages. Both use synthesized speech. For Vietnamese, PhoWhisper (Le et al., 2024) fine-tunes Whisper (Radford et al., 2023); we found no Vietnamese benchmark for spoken function calling.

## 3 Nexa Drive: Task and System

### 3.1 Functions and actions

The assistant controls 24 functions of a simulated car (Table 1; full signatures in Appendix A). Nineteen correspond to properties of the public Android Automotive `VehiclePropertyIds` specification (Google, 2026) (for example `set_ac` ↔ `HVAC_POWER_ON`, `HVAC_TEMPERATURE_SET`; `control_window` ↔ `WINDOW_MOVE`), several properties being merged into one function with a position argument, as in Jafarzadeh Asl et al. (2026). Navigation, nearby search and media follow Google's documentation for Android car apps, and weather follows CAR-bench. Value ranges (temperature 16–30 °C, fan 0–7, volume 0–40) describe one simulated configuration, not a standard. A 25th token, `irrelevant_function`, means "refuse".

For every utterance the model chooses one of three actions:

- **Execute** — *"Hạ kính ghế phụ một nửa"* (*lower the passenger window halfway*) → `<nexa_15>(window="front_right", position="down", amount=50)<nexa_end>`.
- **Clarify** — *"Ôi tôi nóng quá"* (*oh, I am so hot*) → `<nexa_ask><nexa_6>(state="on")<nexa_end>`; the runtime asks "Shall I turn on the air conditioning?" and executes the stored call if the driver says yes, without calling the model again.
- **Refuse** — *"Đặt giúp tôi vé xem phim"* (*book me a movie ticket*) → `<nexa_24>()<nexa_end>`.

| Group | Functions |
|---|---|
| Status (read-only) | fuel level, battery range, tire pressure, vehicle status, odometer, range estimate, weather |
| Climate | AC (on/off, temperature, delta, mode, zone), fan speed, defrost |
| Access | lock, unlock*, open trunk*, close trunk, open charge port*, open fuel flap* |
| Windows | side windows, sunroof |
| Lights | headlights, hazard lights |
| Media, route | volume, play media, navigate, find nearby |

Table 1: The 24 functions. *Sensitive functions (they open access to the car): never proposed from an implicit request and always confirmed when triggered by the wake word.

Independent requests in one call are separated by `;`. Data policy fixed early in the project: read-only functions are never "clarified" (reading a value cannot do harm, asking only annoys the driver); a clarification keeps only arguments the user actually stated; sensitive functions and the high beam are never proposed from an implicit request.

### 3.2 Functional tokens and prompts

We add 27 tokens to the Qwen2.5 tokenizer (Qwen Team, 2024): `<nexa_0>`...`<nexa_24>`, `<nexa_end>` and `<nexa_ask>` (vocabulary 151,665 → 151,692); new embeddings are initialized to the mean of existing ones plus noise. Because the student is an instruction-tuned model, prompts use its native ChatML template. The call prompt is a one-line system message ("You are an agent that controls a car", in Vietnamese) and the user utterance; it averages 33–39 tokens. From data round v6 on, the user turn may start with `Lượt trước: <previous call>` ("previous turn"), which lets the model resolve follow-ups such as *"đóng đi"* (*close it*). As in Octopus v2, we tested appending a natural-language description of the function after `<nexa_end>` (`Mô tả hàm: ...`, "function description"; §6.2). The final training set keeps it in only 2,385 of its 33,523 examples (7.1%, refusals inherited from earlier rounds); at inference the model stops at `<nexa_end>`, so the description costs no latency. A second prompt type (the call, the function result and the user turn) trains the same model to phrase the answer; examples are in Appendix B.

### 3.3 Runtime pipeline

Figure 1 shows the deployed pipeline. The backend (FastAPI + WebSocket) keeps one simulated vehicle and one pending call per session. The student runs in `llama-server` from llama.cpp (Gerganov et al., 2023); prompts are sent as token ids produced by the Hugging Face tokenizer and outputs are decoded from token ids, because the text field of the server hides control tokens such as `<nexa_i>`. The end-of-sequence token is overridden to `<nexa_end>`: without it, uncertain refusals continued into the description seen in those 7.1% of examples until the generation limit, which made them five to eight times slower (identical outputs on 400 validation prompts, total time 540 s → 249 s).

Several small layers sit around the model; each was added after a measured failure (details in Appendix E):

- **Text normalization.** The training data, written by the teacher, always starts with a capital letter. For *"hỏng xe giữa đường rồi"* (*my car broke down in the middle of the road*), the lowercase version is refused (refusal probability 0.68) while the capitalized one triggers a hazard-light proposal (0.99). We therefore capitalize the first letter, collapse repeated letters and lower-case all-caps input before the model sees it.
- **Retry without context.** If the model refuses while a previous turn is in the prompt, it is called once more without the context, so that context can help a follow-up but never turns an understood request into a refusal.
- **Relative-command conventions.** "Colder" without a number means $-2\,^{\circ}$C, fan $\pm 1$ level, volume $\pm 3$—the same rules used to label the training data.
- **Multi-command splitting.** The training data contain no example that executes several independent actions, so the model picks one action from *"bật đèn pha lên xong kiểm tra xăng xong đó mở cốp"* (*turn on the headlights, then check the fuel, then open the trunk*). We split such utterances on explicit sequential connectors (*xong*, *sau đó*, *rồi thì*, …; not on *và* "and"), run the model per segment and stop at the first segment that needs clarification.
- **Answer templates.** The model was trained to phrase answers from function results, but this second pass adds about 1.6 s and produced semantic errors no automatic check caught (§6.6);

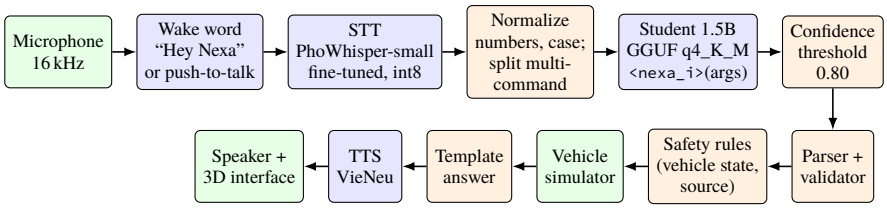



Figure 1: Runtime pipeline. Blue: learned models; orange: code; green: input/output. Clarifications and refusals leave the pipeline after the validator or safety step with a question or an apology; a pending call is executed when the driver confirms. Everything runs on the laptop CPU; the GPU is not used.

the deployed system answers with templates filled from the result.

- **Closing the fuel flap and charge port.** The schema can open these flaps but has no function that closes them (many cars close them by hand), so the model maps *"đóng nắp xăng"* (*close the fuel flap*) to the nearest function, which *opens* it. A runtime rule recognizes a closing verb before the flap name, or *"đóng đi"* (*close it*) right after one was opened, and closes the flap without calling the model; the interface marks the action as rule-based. Unusual phrasings and input without diacritics are not covered; the principled fix is a closing function and another training round.

| Condition | Action |
|---|---|
| open trunk / charge port / fuel flap while driving | refuse |
| unlock doors while driving | ask to confirm |
| any sensitive function via wake word | ask to confirm |
| child lock on + wake word + rear window down or sunroof open/tilt | ask to confirm |
| driving starts with trunk, door or flap open | warn |
| model proposes a sensitive function | drop the proposal |

Table 2: Runtime safety rules (code, independent of the model).

### 3.4 Guardrails

Following the principle "the model understands, code keeps the car safe", every call passes a validator and a set of state-dependent rules declared in the function schema (Table 2). The validator rejects values outside enumerations or ranges, asks for missing required arguments and drops unknown arguments. Each command carries its source: push-to-talk is treated as the driver, while the wake word can be spoken by any passenger. Mirroring a real window-lock button, the child lock lets the driver lower rear windows directly but asks for confirmation when the request comes through the wake word. The rules are covered by 963 automated tests, including scenario tests that, as in $\tau$-bench, check the final vehicle state rather than the wording of the answer. We did not measure guardrails with a quantitative benchmark.

## 4 Data

### 4.1 Distillation from a local teacher

The teacher is Qwen2.5-72B-Instruct (AWQ 4-bit, Lin et al., 2024a) served with vLLM (Kwon et al., 2023) on a rented A100 80 GB; no paid API is used. For each function the teacher writes Vietnamese requests in batches of 50, extracts the call arguments in a second pass, writes natural answers for a sample of calls given a simulated function result, and writes out-of-scope requests that should be refused, keeping positives and refusals at about 1:1. The first pool held 27,327 samples (about 1,000 per function) and cost about US$13–15 of GPU time. Data is split into train and validation by groups of near-duplicates (MinHash with a fixed hash, so the split is reproducible), so that paraphrases of the same sentence never cross the split.

### 4.2 Quality control

We follow the rule "whatever code can check, code checks" (Liu et al., 2024b,a). Code removes arguments outside enumerations or ranges, Chinese/Japanese/Korean characters (119 samples in the first pool), leaked function names, invented numbers in answers, mixed-English sentences and commands hidden in clarification samples. Failing samples are discarded, not regenerated: regenerating until a judge agrees moves the judge's bias into the data.

**Judges.** The teacher also judges its own data, which checks consistency rather than correctness. For clarification samples we therefore use a jury (Verga et al., 2024): Qwen2.5-72B and Llama-3.3-

70B (Grattafiori et al., 2024) each pick up to two functions for the sentence *without seeing the label* (Zheng et al., 2023), and a sample is kept only if both picks contain the label. Every judging job starts with a labeled calibration set and stops if agreement in either direction is below 80% or Cohen's $\kappa$ below 0.6 (Thakur et al., 2025); the jury reached $\kappa = 0.875$. The previous single-judge prompt had kept 517 of 518 clarification samples, including clearly wrong ones such as "this parking spot looks unsafe" $\rightarrow$ hazard lights.

**Human audits.** Against 400 reference labels (drafted by an LLM and corrected by the first author), the two judges agree with the reference on the first function only 64–70% of the time. The jury filter raises label precision from 75% to 88% while wrongly discarding 16% of correct samples. Two additional code rules (imperative particles at the end of a clause; hazard lights of *another* car), designed on one half of the 400 labels, raise precision on the other half from 86.2% to 90.8%. A stratified audit of 195 call/answer/refusal samples by the first author found 90.3% correct (95% Wilson CI 85–94), below APIGen's 95.3%, and exposed three systematic errors that machine judges had accepted: "is the fuel enough for $N$ km" labeled as a fuel-level query (21% of that function), "volume up to $N$" labeled as a relative change (26%), and window direction inverted for "close". They were fixed by code rules in round R3. A second audit of 52 samples from round v5 found 90.4% correct. The lesson we drew: two machines agreeing is not evidence of correctness—Llama-3.3-70B accepted 95.2% of the samples Qwen had approved, while a person found systematic errors in under an hour.

### 4.3 Evaluation sets

No benchmark covers Vietnamese in-car function calling with functional tokens, so we built evaluation sets that are separate from the training pipeline in source (Table 3). Sentences were generated with GPT6-Astra (a web-based LLM assistant) from a written description of what the assistant can and cannot do—no example sentences and no training data were shown—and labeled by GPT6-Astra in a separate conversation from a lookup table; the first author then reviewed and corrected every label. Using a different generator from the teacher avoids the setting of Octopus v2, where training and test queries come from the same model. We call these sets *synthetic, human-reviewed*; they are not human-written. Every set is checked against all training and validation data and every few-shot example in the schema (MinHash near-duplicates), and frozen with a SHA-256 hash in the test suite before any model is scored on it. The original 296-sentence test set lost 66 sentences to contamination (short car commands are often identical), leaving the 230-sentence *clean test* used throughout.

| Set | $n$ | Purpose |
|---|---|---|
| Clean test | 230 | 74 clear, 39 ambiguous, 117 out-of-scope |
| Supplement | 28 | complaints for 7 functions |
| Ambig-3 | 130 | implicit requests, 13 functions; used once |
| Ambig-4 | 129 | implicit, mostly long requests |
| v6-A / B / C | 150 | short (72), paraphrase/dialect (48), new arguments (30) |
| v6-D | 60 | two-turn follow-ups |
| v6-E | 104 | no diacritics or typos (code) |
| GRPO test | 197 | 4 parts, plus a lower-case copy |
| Calibration | 194 | hard out-of-scope, for thresholds |
| Dev | 30 | story-like requests, 20 refuse |
| Validation | 3,416 | teacher-generated; 600 for checkpoints |

Table 3: Evaluation sets. All except validation and v6-E are generated by GPT6-Astra from a written specification and reviewed label by label by the first author.

### 4.4 Data rounds

The training data went through six rounds; each round changed one group of factors, was diagnosed only on training statistics and validation errors (never by reading failed test items), and was accepted or rejected by criteria written before training. Table 4 summarizes the rounds; §6.2 reports what they taught and Appendix C gives sizes, costs and details.

The final training set (v6.1) has 33,523 examples: 28,635 call targets (of which 13,778 are refusals), 3,158 clarifications and 1,730 answer-phrasing examples; 4,686 carry a previous-turn context and about 11% (3.6k) contain removed diacritics or typos. Refusals make up 43.3% of call and clarification targets. Examples average 54 tokens and none exceed the 256-token limit.

## 5 Training and Evaluation Protocol

### 5.1 Training

The student is Qwen2.5-1.5B-Instruct. We use the plain Hugging Face `Trainer` with the loss

| Round | Main change | Train |
|---|---|---|
| R1 | teacher pool, 1:1 refusals, balanced 300/function | 15.9k |
| R2 | synonyms per function, imperative refusals, jury for clarifications | 16.1k |
| R3 | code fixes for 3 systematic label errors | 16.0k |
| R4 | story-like refusals (rejected) + contrastive story-like calls | 17.6k |
| v5 | story-like clarifications (third contrastive pair) | 19.3k |
| v6 | short/dialect phrasing, AC delta & mode, both rear windows, previous-turn context, typos | 28.8k |
| v6.1 | v6 + 3.3k existing refusals to restore the refusal ratio | 33.5k |

Table 4: Data rounds. Side experiments (function descriptions in targets, 1,000 samples per function, a combined configuration) are in §6.2.

masked by the token length of the prompt, rather than TRL's `SFTTrainer`, whose completion-only collator searches for the assistant marker (which appears twice in answer-phrasing samples) and silently masks a whole sample when it does not match. Hyperparameters follow Octopus v2: AdamW, learning rate $5 \times 10^{-5}$, 10 warm-up steps, linear decay, 3 epochs, equal loss weights for the new tokens (Octopus v2 found weighting unnecessary for few tokens), effective batch 16, maximum length 256, bf16. LoRA uses rank 16, $\alpha = 32$ on `q,k,v,o,up,down` projections; because the new tokens must be learned, `embed_tokens` and `lm_head` are trained in full. Qwen2.5-1.5B ties these two matrices; saving a merged PEFT model with tied weights silently discards the trained output layer, so we untie them before training. With the two full matrices, 480M of 2.26B parameters are trainable (21%). Each configuration is trained with seeds 42 and 7 on a rented A100 40 GB, L40S or L4 (v6.1: about 57 minutes and 6,288 steps per seed). A checkpoint is saved per epoch and chosen on 600 stratified validation samples: the highest accuracy on clear commands among checkpoints with false refusals $\leq 5\%$, earlier epoch on ties.

### 5.2 Scoring

All models, including prompted baselines whose JSON output is converted to the functional format, are scored by the same code. An output is *correct* if it matches the reference after normalization ("rule C"): arguments equal to their schema default are dropped on both sides (`get_odometer()` = `get_odometer(unit="km")`); for ambiguous items only the function and the direction are checked (on/off, open/close, up/down, sign of a delta), and a proposal may pick any one of several acceptable functions; suggesting the high beam is always wrong. Real behavioral differences (delta 1 vs. 2, miles vs. kilometres) are never forgiven. Rule C was fixed before round R2 and old results were re-scored with it; under the earlier exact-match rule the same R1 outputs score 4.1 points lower on clear commands.

Errors are classified into seven types: wrong function, wrong argument, false refusal, over-triggering (calling a real function on an out-of-scope request), unnecessary clarification, missing clarification, and unparseable output. We call over-triggering, wrong function, wrong argument and missing clarification *wrong actions*: the car would do something the user did not ask for. We report per-group accuracy and never a single pooled number: a model that refuses everything already scores 51% on the clean test.

### 5.3 Statistical protocol

The test sets are small (74 clear commands; one item is 1.35 points), so we use Wilson intervals (Wilson, 1927) and the exact two-sided McNemar test (McNemar, 1947) on paired outputs. McNemar accounts for the choice of test items but not for training randomness: when we retrained round R2 with a second seed on identical data, 33 of 230 items changed outcome and McNemar between the two seeds gave $p = 0.005$. From then on every configuration was trained with two seeds (three for the final one, Appendix D), results are the mean of the two, and a difference is claimed only if the mean difference exceeds the spread between seeds and, for paired tests, if it is significant for both seeds in the same direction (one seed: "tie or slightly better"). The criteria for accepting each round, choosing thresholds and compression levels were recorded before the corresponding test run, and we did not change them after seeing scores. Two seeds remain few; our conclusions are correspondingly modest.

## 6 Results

### 6.1 Student versus prompted baselines

Table 5 compares the student with schema-in-prompt (SIP) baselines: the same prompt—descriptions of the functions, the three actions, the same data policy and three examples, about 2,150 tokens—given to the untrained Qwen2.5-

1.5B-Instruct and to the 72B teacher, each run once with greedy decoding.

**Training is necessary.** With the same 1.5B model, prompting reaches only 14–38% on every group (21–28% on the newer short, two-turn, typo and Ambig-4 sets, Table 8); the 1.5B prompted model over-triggers on 63% of out-of-scope requests and returns invalid JSON for 10 clean-test items. Fine-tuning with functional tokens roughly doubles to quadruples every group while the prompt is 65 times shorter.

**Against the 72B teacher.** Exact McNemar tests per seed (v6.1, unquantized, Appendix D) give: *clear commands* 81.1 and 75.7 vs. 83.8 ($p = 0.77$, $0.15$): no detectable difference; *long implicit requests* (Ambig-4) 79.8 and 78.3 vs. 70.5 ($p = 0.065$, $0.14$): higher but not significant; the small ambiguous group: significant for one seed only (tie or slightly better); *out-of-scope* 82.9 and 71.8 vs. 94.9 ($p \leq 0.001$ for both): the teacher is clearly better. On the v6 sets (Table 8; the prompted baselines were run on them later with the same prompt and decoding) the picture is similar: *short commands and paraphrases* 88.3 and 87.5 vs. 91.7 ($p = 0.39$, $0.27$) and *typos* 60.6 and 66.3 vs. 69.2 ($p = 0.093$, $0.66$) are ties, *new arguments* favor the student (83.3 vs. 70.0, $p = 0.34$), but on *two-turn follow-ups* the teacher is clearly better (81.7 vs. 66.7 and 58.3; $p = 0.049$, $0.001$), although it never saw a two-turn example and receives exactly the previous call that the student receives. Part of the 72B's errors on clear commands are label conventions it cannot know from descriptions (an AC temperature without `state="on"`, "weak lights" $\rightarrow$ high beam). A 1.5B model can tie a 72B one on clear and implicit requests here because the comparison is not symmetric: the student learned this task's conventions from 33k examples of the teacher's *filtered* outputs, whereas the teacher works zero-shot from a prompt; where general knowledge or reasoning over context is needed—recognizing what the car cannot do, or how a follow-up modifies the previous action—the larger model remains ahead.

### 6.2 What the data rounds taught

Figure 2 traces clean-test accuracy over the configurations we trained and Table 6 gives the side experiments on top of R3. Five findings shaped the final data.

**Validation is not a test.** From R1 on, validation accuracy on clear commands was 97–99%, while the clean test gave 65–80%. Validation data come from the same teacher and prompts as training data, so they share its blind spots; only the separately generated test sets revealed them.

**Lexical lock-in.** In R1, 15 of 24 functions had one word covering $\geq$ 95% of their training sentences. All 267 window sentences said *cửa sổ* (window) and none said *kính* (glass), while every defrost sentence said *kính*; the model learned "*kính* $\Rightarrow$ defrost" and turned "*Hạ kính ghế phụ*" (*lower the passenger glass*) into `set_defrost(state="down")`. Asking the teacher to spread synonyms evenly (R2) fixed this: all nine *kính* window commands in the clean test now chose the window function, and window accuracy rose from 30% to 80%.

**Shortcuts from sentence form.** R1 refusals were almost never imperative (2.1%), so "imperative $\Rightarrow$ call" was 94.7% accurate on the training data. Adding 1,667 imperative refusals in R2 broke this rule (60.4%) but did not reduce over-triggering (18.7% $\rightarrow$ 17.8%). In R4 we generated 1,651 refusals that first describe a situation and then ask for something the car cannot do. Development-set refusals jumped from 8.5/20 to 20/20, but story-like *doable* requests were now refused (2–6 of 10, vs. 10/10 before): the model had learned "story $\Rightarrow$ refuse". Both R4 variants were rejected by their pre-registered false-refusal criterion. Adding the same story frame on the call side (1,641 story-like calls, the *contrastive* configuration) removed the shortcut and gave the largest out-of-scope gain of the project (57.2 $\rightarrow$ 74.8, with 10/10 doable requests kept). We now generate hard examples only in pairs across labels.

**The same lesson for clarifications.** On a new 130-item ambiguous set (Ambig-3) with a pre-registered test, the prompted 72B beat the contrastive student clearly (71.5 vs. 51.5 and 38.5; $p < 0.001$ for both seeds). The student collapsed on long, story-like complaints, mostly by refusing them. Training data explained why: among sentences longer than ten words there were 3,254 refusals but only 102 clarifications. Round v5 added 1,696 story-like clarifications (a third contrastive pair), judged by the jury and audited (90.4%). On a fresh set, Ambig-4, v5 reached 74.4 and 72.1 vs. 51.2 and 45.0 for the contrastive model ($p < 10^{-4}$,

| System | Clean test (230) | | | Suppl. | Ambig-4 | Prompt | p50 |
|---|---|---|---|---|---|---|---|
| | Clear (74) | Amb. (39) | OOS (117) | (28) | (129) | tokens | latency |
| Qwen2.5-1.5B, prompted (SIP) | 37.8 | 35.9 | 31.6 | 14.3 | 20.9 | 2,156 | 779 ms |
| Qwen2.5-72B teacher, prompted (SIP) | **83.8** | 56.4 | **94.9** | 39.3 | 70.5 | 2,156 | 1,842 ms |
| Student v5, LoRA | 78.4 | 75.6 | 83.8 | 58.9 | 78.3 | 33 | |
| Student v6.1, LoRA | 78.4 | 75.6 | 77.4 | 58.9 | **79.1** | 33 | 224 ms[†] |
| Student v6.1 + threshold 0.85 | 75.0 | 74.4 | 82.1 | 58.9 | **79.1** | 33 | |
| Deployed: v6.1 s42, 4-bit, $\tau$ 0.80 | 77.0 | 64.1 | 88.0 | 50.0 | 74.4 | 33 | 463 ms[‡] |

Table 5: Accuracy (%) on the frozen evaluation sets. Student rows are means of two seeds, unquantized; baselines are single greedy runs; OOS = out-of-scope. Latency: A100 with vLLM, one query at a time ([†]measured for an earlier student with the same prompt length); [‡]laptop CPU, llama.cpp (520 ms in the first scoring session, §6.6). The v5 row comes from the same re-generation as v6.1 and differs by at most two items per group from the original v5 run in Table 7. The deployed row is one seed after 4-bit quantization with the threshold (§6.6).

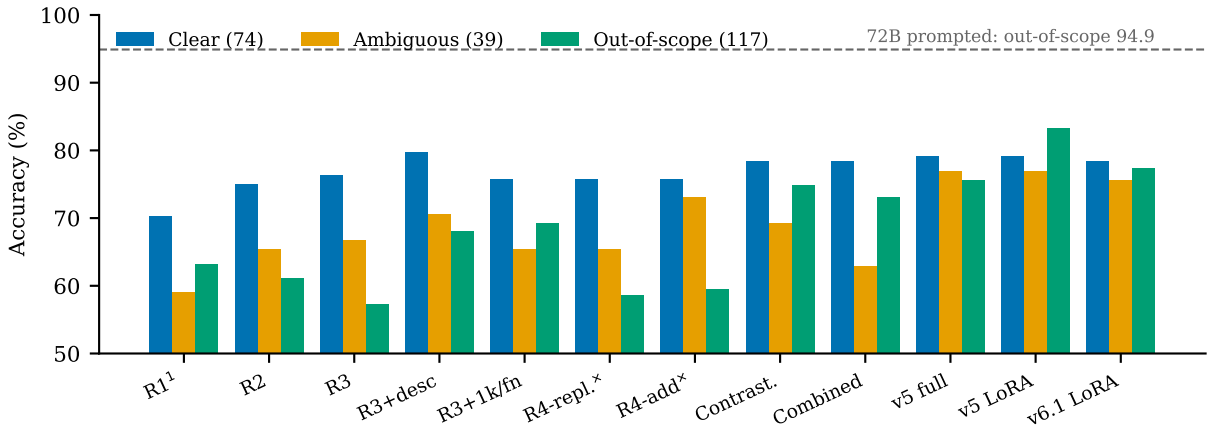



Figure 2: Clean-test accuracy across data configurations (means of two seeds; [1]single seed; [×]rejected by pre-registered criteria because they refused doable requests). The v6.1 row is from a later re-run of the same scorer.

both seeds) and became indistinguishable from the 72B (70.5; $p = 0.51$, 0.87); false refusals on long complaints fell from 18 and 32 to 6 and 6, the 72B's level.

**Octopus-style descriptions and data scale.** Appending function descriptions to targets raised clear commands by 3.4 points with almost no seed spread; using up to 1,000 samples per function (all of the pool) raised out-of-scope accuracy by 12 points. Combining both with the contrastive data brought no further gain: data scale and contrastive refusals fix the same weakness, and the description effect was diluted (Table 6). We kept the smaller contrastive set (17.6k vs. 30.0k examples) as the base for v5.

### 6.3 LoRA versus full fine-tuning

Octopus v2 reports full fine-tuning slightly above LoRA (99.524% vs. 99.048%, which corresponds to one item if their test has 210 queries). On the v5 data, with the same learning rate, seeds and selection rule, LoRA matches full fine-tuning on clear commands and is better on out-of-scope requests (Table 7). The out-of-scope advantage holds in all four comparisons at equal epochs (two epochs $\times$ two seeds, +5 to +13 points); the advantage on Ambig-4 disappears at equal epochs, so we attribute it to checkpoint choice. Full fine-tuning showed signs of overfitting (validation loss rising at epoch 3, over-triggering growing with epochs).

| Config. (vs. R3) | Clear | Amb. | OOS | Dev ref. | Dev do |
|---|---|---|---|---|---|
| R3 | 76.3 | 66.7 | 57.2 | 8.5 | 10 |
| + descriptions | **79.7** | 70.5 | 68.0 | 7.0 | 10 |
| + 1,000 / function | 75.7 | 65.4 | 69.2 | 10.0 | 10 |
| R4 replace[×] | 75.7 | 65.4 | 58.6 | 20.0 | 6 |
| R4 add[×] | 75.7 | 73.1 | 59.4 | 20.0 | 2 |
| Contrastive | 78.4 | 69.2 | **74.8** | 17.5 | 10 |
| Combined | 78.3 | 62.8 | 73.0 | 16.5 | 10 |

Table 6: Data experiments on top of R3 (full fine-tuning, means of two seeds). Dev ref.: refusals correct out of 20; Dev do: doable story-like requests executed out of 10. Seed spreads are in Appendix C.

| v5 data, mean of 2 seeds | Full 5e−5 | Full 2e−5 | LoRA |
|---|---|---|---|
| Clear (74) | 79.1 | 71.6 | 79.1 |
| Ambiguous (39) | 76.9 | 48.7 | 76.9 |
| Out-of-scope (117) | 75.6 | 76.1 | **83.3** |
| Ambig-4 (129) | 73.3 | 46.5 | 78.3 |
| False refusal | 0.7 | 0.4 | 0.2 |

Table 7: Full fine-tuning vs. LoRA (r 16, $\alpha$ 32) on identical data. McNemar on the clean test: $p = 0.019$ (seed 42), 0.25 (seed 7).

Lowering its learning rate to $2 \times 10^{-5}$ did not close the gap (76.1 on out-of-scope; pre-registered verdict "still behind") and left the model under-trained on everything else. We therefore deploy LoRA, but do not claim LoRA is better in general.

### 6.4 Short commands, context and typos (v6, v6.1)

Testing the v5 demo revealed failures that the earlier test sets did not contain: terse commands (*"bật pha"* (*high beam on*) was refused), dialect words (*kiếng* for *kính*), missing diacritics (*"mo cop"* (*open trunk*)), follow-ups (*"đóng đi"* (*close it*) after opening the trunk), relative temperature (*"giảm nhiệt độ điều hòa"* (*lower the AC temperature*) turned the AC *off*, because the schema had no relative argument), and both rear windows at once. Measuring the v5 training data confirmed that teacher sentences are long and formal: only 0.7% of call sentences have three words or fewer, and 14 functions have none. Round v6 extended the schema (AC `delta` and `mode`, fan `delta`, windows `front/rear`) and generated data from assigned combinations of verb, part name, abbreviation, dialect, length tier and register, plus two-turn data with the previous call in the prompt and typos introduced by code. A new 210-item test was written and frozen before any v6 model existed.

Table 8 shows a large gain on everything new and an out-of-scope regression. The training data explained it: v6 added about 9,500 mostly positive examples and the refusal share fell from 40.5% to 33.3%. v6.1 restored it to 43.3% with 3,300 refusals that had already passed earlier filters (no new generation), which halved the regression ($-10.7 \rightarrow -6.4$) but still failed criterion (4), mainly because of seed 7 ($-11.1$; seed 42 $-1.7$). By our rule v6.1 does not replace v5 as the *method* result, and we report both. For the demo we nevertheless chose v6.1 seed 42, because it handles the commands drivers actually produced in testing; seed 42 was chosen on the independent calibration set (§6.5), not on test scores. The rule-based memory we had written before v6 barely helped on real follow-ups (23.3, same as no memory): human follow-ups are rarely a bare verb, whereas the model with context resolved 80–87% of direction changes.

| | Student (2 seeds) | | | | Prompted (SIP) | |
|---|---|---|---|---|---|---|
| | v5 | v6 | v6.1 | crit. | 1.5B | 72B |
| (1) Short + paraphrase (120) | 76.3 | 88.8 | 87.9 | pass | 27.5 | **91.7** |
| (2) Two-turn (60)[a] | 23.3 | 60.8 | 62.5 | | 28.3 | **81.7** |
| (3) New arguments (30) | 8.4 | **83.3** | **83.3** | | 16.7 | 70.0 |
| (4) Clear, old test (74) | 78.4 | 77.7 | 78.4 | pass | 37.8 | **83.8** |
| (4) Out-of-scope (117) | 83.8 | 73.1 | 77.4 | fail | 31.6 | **94.9** |
| (4) Ambig-4 (129) | 78.3 | **80.6** | 79.1 | pass | 20.9 | 70.5 |
| (5) Typos (104) | 31.7 | 64.4 | 63.5 | | 21.2 | **69.2** |

Table 8: Pre-registered comparison of the student rounds (criteria compare v6.1 with v5 only), with the prompted baselines for reference (single greedy runs; bold = best per row). Criterion (1): better than v5 for both seeds (McNemar $p = 0.0007$, 0.007 for v6.1); (4): no group of the old tests more than 3 points below v5. [a]v5 with a rule-based memory (without memory also 23.3); prompted baselines receive the previous call in the user message, as the student does.

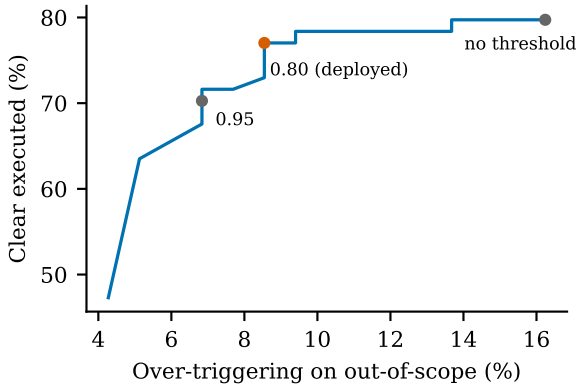



Figure 3: Effect of the threshold for the deployed model on the clean test (descriptive; the threshold was chosen on the calibration set).

### 6.5 Confidence threshold

With functional tokens, the probability of the first generated token is the probability of the decision. Our rule: if the top token is a function with $p < \tau$, refuse when the second token is the refusal token, otherwise ask the user to confirm the full top call (so "yes" executes it); refusals and clarifications chosen by the model are left alone. Thresholds are chosen on non-test data and then fixed.

On the clean test the threshold consistently trades severe errors for mild ones. For the contrastive model ($\tau = 0.90$, chosen on validation)

wrong actions fell from 53 to 33 and from 44 to 25 per seed (about $-40\%$); for LoRA v5 ($\tau = 0.85$) from 34/38 to 21/23; for the deployed model ($\tau = 0.80$) from 41 to 29, with over-triggering on out-of-scope requests falling from 16.2% to 8.5% (Figure 3). The price is extra confirmations: clear-command accuracy drops (e.g. LoRA v5 79.1 $\rightarrow$ 70.9), but every clear command lost this way was replaced by a confirmation proposing the correct call, so the share solved after at most one "yes" is unchanged (59 = 52 + 7 items for seed 42). Thresholds must be re-chosen per model: LoRA is less confident than full fine-tuning, 4-bit quantization lowers confidence further, and an under-trained model lost every clear command at $\tau = 0.90$.

Our first rule (largest $\tau$ that loses $\leq 1\%$ of clear commands on validation) was arbitrary, and validation contains too few over-triggers to choose from (3–14). We therefore built a 194-sentence calibration set of hard out-of-scope requests and asked for a guarantee: the lowest $\tau$ whose one-sided 90% Clopper–Pearson bound (Clopper and Pearson, 1934) on wrong behavior is $\leq 10\%$. *No model met it at any* $\tau \leq 0.99$: 19–28% of these requests are handled wrongly without a threshold and 9–16% remain at 0.99, because the model is confidently wrong. Separating the two kinds of error is informative: self-executed wrong actions fall from 9–13% to 3–5% at $\tau = 0.85$, and what the threshold cannot touch are clarification proposals in response to chit-chat ("seeing these flame trees reminds me of school" $\rightarrow$ "open the sunroof?"), a side effect of the story-like clarification data that is present in v5 and v6.1 alike. The deployed threshold 0.80 was chosen on this set (5 self-executed errors out of 194, as at 0.85, with fewer lost commands).

### 6.6 Quantization and deployment

The model is converted to GGUF and quantized with llama.cpp. The compression level was chosen by a rule fixed in advance on validation and development data (3,036 items): accept the first of q4_K_M, q4_K_M+imatrix, q5_K_M, q8_0 that loses at most 0.5% of correct clear commands, adds at most two over-triggers and loses at most two ambiguous items relative to the unquantized model. For v5, f16 conversion changed 5 of 3,036 outputs, q8_0 5, q5_K_M 24 and q4_K_M 38; q4_K_M passed (q5_K_M failed by one ambiguous item), so q4_K_M was chosen (Table 9). llama.cpp keeps the output layer at 6 bits even in q4_K_M, which protects the rows of the 27 new tokens. The f16 conversion on the same CPU reproduced the unquantized model, so the remaining differences come from rounding, not from the hardware.

| v5, val+dev | Size | Clear | Over-trig. | Amb. | Same |
|---|---|---|---|---|---|
| HF bf16 | 3.31 GB | 1,506 | 14 | 127 | – |
| f16 | 3.31 GB | 1,508 | 14 | 126 | 3,031 |
| q8_0 | 1.76 GB | 1,507 | 15 | 126 | 3,031 |
| q5_K_M | 1.20 GB | 1,505 | 13 | 124 | 3,012 |
| q4_K_M | 1.04 GB | 1,503 | 12 | 125 | 2,998 |

Table 9: Quantization of the v5 model on 3,036 validation and development items: correct clear commands, over-triggers, correct ambiguous items and outputs identical to HF.

**Deployment on test.** After freezing everything, we scored the deployed v6.1 model once on all frozen sets. On validation, q4_K_M had again passed the rule (+4 clear, +0 over-triggers, $-2$ ambiguous, 2,971 of 3,006 outputs identical), yet on test it scored below the unquantized model run on GPU, most visibly on long implicit requests (Ambig-4 79.8 $\rightarrow$ 74.4) and short commands (88.3 $\rightarrow$ 82.5). To separate the effect of rounding from that of changing the inference stack, we also converted the same model to f16 and q8_0 GGUF and scored f16, q8_0 and q4_K_M one after another on the same CPU (Table 10; re-scoring q4_K_M reproduced all 701 earlier outputs). Two effects appear. First, moving from batched bf16 inference on GPU to f16 on CPU already flips near-boundary items: on the small ambiguous group f16 scores 69.2 against 74.4. Second, relative to f16 on the same CPU, q4_K_M is significantly worse on short commands (0 vs. 7 flipped items, $p = 0.016$) and close to significant on Ambig-4 (1 vs. 7, $p = 0.070$), whereas q8_0 gives the same output as f16 for 680 of 701 prompts and differs significantly on no group. Validation items are confident (top-token probability near 0.99), so small numeric shifts rarely flip them; the human-reviewed sets contain many near-boundary decisions that do. We keep q4_K_M because the choice was made by a fixed rule on non-test data, and note that this rule is not sensitive enough to distribution shift. q8_0 is the safer choice on stronger hardware: it is $1.7\times$ larger and, in the same session, $1.8\times$ slower than q4_K_M (median 856 vs. 463 ms). With the threshold, all three levels make the same number of wrong actions on the clean test (29). On the

| v6.1 seed 42 | HF (GPU) | f16 | q8_0 | q4 | q4 + 0.80 |
|---|---|---|---|---|---|
| Clear (74) | 81.1 | 81.1 | 81.1 | 79.7 | 77.0 |
| Ambiguous (39) | 74.4 | 69.2 | 69.2 | 64.1 | 64.1 |
| Out-of-scope (117) | 82.9 | 82.1 | 82.1 | 82.9 | 88.0 |
| Ambig-4 (129) | 79.8 | 79.1 | 78.3 | 74.4 | 74.4 |
| Short + paraphr. (120) | 88.3 | 88.3 | 85.0 | 82.5 | 79.2 |
| Two-turn (60) | 66.7 | 66.7 | 65.0 | 61.7 | 56.7 |
| Typos (104) | 60.6 | 58.7 | 60.6 | 55.8 | 47.1 |
| Wrong actions, clean | 37 | 40 | 40 | 41 | 29 |
| Median latency (CPU) | – | 1,062 | 856 | 463 | 463 |

Table 10: The deployed model (v6.1 seed 42) on the frozen test sets. HF: unquantized, batched bf16 on GPU. The other columns are GGUF on the same laptop CPU in one session; latency in ms per decision (p95: 2,075 / 1,480 / 832). The threshold does not change the model's own clarifications, so the ambiguous rows of the last two columns are identical.

laptop CPU (i5-12500H, 8 threads) the deployed model needed 520 ms at the median and 961 ms at the 95th percentile per decision in the first scoring session (463 and 832 ms in the second). On the clean test the deployed model makes 29 wrong actions on the car, the prompted 72B teacher 32 and the prompted 1.5B model 95; we did not test the difference between 29 and 32 for significance.

**Answer phrasing.** The model can also phrase the answer from the function result (second prompt). Measured with the v5 model on 200 validation samples, it refused correctly in 99 of 99 out-of-scope samples, and 87.1% of informative answers passed the code checks (no invented number, no English words), but it also produced semantic errors that no check caught ("today" for a two-day forecast, a wrong pressure comparison) and added a median 1.6 s. The deployed system therefore answers with templates; model answers remain available as an option.

### 6.7 GRPO after distillation

Since the remaining errors are decisions (call, clarify or refuse), we tried GRPO (Shao et al., 2024) with a code-computed reward that weights errors by their consequence: correct $+1$, unnecessary clarification $-0.2$, false refusal or wrong proposal $-0.5$, unparseable $-1$, wrong action on the car $-5$. The magnitude $-5$ is tied to the runtime threshold: acting at $p = 0.8$ has the same expected reward as asking ($0.8 \cdot 1 - 0.2 \cdot 5 = -0.2$). A fresh LoRA (r 16) on the merged v6.1 model was trained with TRL, $G = 8$, learning rate $10^{-5}$, KL coefficient 0.04 and temperature 1.0 for two epochs on about 4,000 training prompts on which v6.1 was uncertain ($p_{\text{top}} < 0.95$, including lower-cased copies) plus 400 confident ones. A new 200-item test set (197 after removing 3 contaminated items, plus a lower-cased copy) was frozen before training, together with five acceptance criteria.

| Criterion (mean of 2 seeds) | v6.1 | GRPO | |
|---|---|---|---|
| (1) Wrong actions ↓ | **6.3%** | 6.9% | fail |
| (2) Implicit (M) | 80.1 | 81.2 | pass |
| (3) Clear (R) | 85.0 | 85.0 | pass |
| (4) False refusals ↓ | 1.3% | **0.6%** | pass |
| (5) Unneeded clarif. in R | 0.0 | 0.0 | pass |

Table 11: GRPO vs. its starting point on the GRPO test set (original and lower-cased copies). GRPO had to reduce wrong actions without hurting the other criteria.

GRPO failed the main criterion (Table 11). It barely changed the model: at the end of training 81–87% of the 8-sample groups had identical rewards (no learning signal), the largest weight change in a query projection was $2.4 \times 10^{-4}$, and only 1–16 of 197 items changed outcome per evaluation. Seven items flipped from correct to a wrong action (unsupported features close to real ones, such as "honk the horn" or "lane-departure warning") and three in the other direction, a difference within noise. GRPO did improve what it practiced: 60% of the uncertain prompts were lower-cased copies, and false refusals on lower-cased input fell from 8 to 3 for one seed—a problem the runtime already solves by capitalization. Compared with tool-use RL that succeeds (Qian et al., 2025; Zhang et al., 2025), our setting lacks a reasoning step (the decision is one token, so sampling explores little), and our prompts were already solved by the model, whereas DAPO (Yu et al., 2025) continuously filters such groups. We keep v6.1 and report GRPO as a negative result (cost about US$2.2).

## 7 Speech Interface

Voice is the primary interface; typing is a fallback. Speech recognition (STT) and synthesis (TTS) run in a separate local service. TTS is used off the shelf. STT was first used off the shelf (§7.1–7.2); the deployed recognizer is a version fine-tuned on car commands (§7.3).

### 7.1 Recognition, numbers and end-to-end accuracy

**Setup.** Candidates were Parakeet-CTC-0.6B-Vietnamese (NVIDIA, 2026) (exported to ONNX, int8), PhoWhisper-small and PhoWhisper-

| Set B (40, real voice) | Correct | WER | p50 | p95 |
|---|---|---|---|---|
| Typed (same chain) | 30 | – | – | – |
| Parakeet-CTC 0.6B | 15 | 33.3 | 388 ms | 489 ms |
| PhoWhisper-small | **21** | 20.2 | 1,844 ms | 4,716 ms |
| PhoWhisper-medium | 18 | **17.0** | 4,806 ms | 5,136 ms |

Table 12: STT selection (pre-registered rule: most end-to-end correct; within two items, lower p95 latency). Measured before the number-normalization fix.

medium (Le et al., 2024) (CTranslate2 int8 via faster-whisper), all on 8 CPU threads with greedy decoding. TTS is VieNeu-TTS (pnnbao-ump, 2026) (0.1B, 48 kHz, ONNX on CPU, female northern voice). The recognizers write numbers as words (*hai mươi hai độ*, "twenty-two degrees"), whereas the student was trained on digits, so transcripts go through a normalizer that converts spoken numbers to digits, removes final punctuation and capitalizes the first letter. The end-to-end metric is: speech → STT → normalizer → deployed model → rule-C scoring against the frozen reference, alongside the same chain fed with the original text ("typed"). Set A (230 clean-test sentences read by 25 synthetic voices) served only for debugging; sets B, C and D (40, 30 and 30 sentences sampled from frozen test sets, each list locked before reading) were recorded by the first author with a laptop microphone in a quiet room.

**Choosing the recognizer.** PhoWhisper-small won the pre-registered comparison (Table 12). PhoWhisper-medium had the lowest word error rate but fewer correct commands: WER weighs every word equally, while one wrong number or part name breaks a command (medium heard "25 degrees" as *hai găm độ*). Synthetic speech overrated Parakeet (WER 21.5% on set A, 33.3% on real voice): the CTC model glues or splits syllables on real speech, while the Whisper decoder uses linguistic context. The ranking held on the new sets C and D, scored afterwards for reference only: PhoWhisper-small 21 and 17 of 30 correct commands, PhoWhisper-medium 19 and 15, Parakeet 14 and 12 (Table 20).

**A bug found by real voice.** Of the 11 commands PhoWhisper-small lost on set B, three were caused by our own normalizer: Vietnamese speakers say *hai hai độ* for 22 degrees, and the normalizer added the digits (4 degrees). After the fix, a new set C gained four commands (17 → 21 of 30), all colloquial numbers; set A, read by TTS voices that pronounce numbers in full, did not change at all. Rescoring set B after the fix (22 of 40) is reported only as post-hoc. Vocabulary hints and beam search were then compared on a new set D under a pre-registered rule: greedy without hints scored 17, hints 16, beam 5 17, both 16, so greedy was kept.

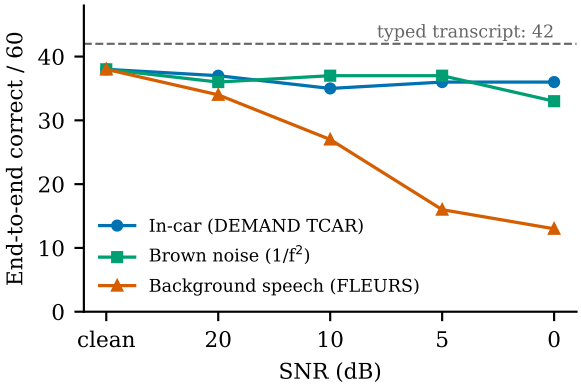



Figure 4: End-to-end correct commands (of 60) under additive noise.

**End-to-end.** The headline numbers for the off-the-shelf recognizer pool the 60 new sentences of sets C and D: **42 of 60 (70.0%) typed and 38 of 60 (63.3%) spoken**, so speech costs 6.7 points. Per set the drop was 22.5 points on B (before the fix), 10 on C and 3.3 on D. For context, SFC-Bench reports drops of 17–20 points for 8B–32B models at its easiest level (Peng et al., 2026); the settings differ in language, functions, metric and speech (synthetic there, one real speaker here), so we place our numbers in that range rather than compare them. The remaining speech errors are real mishearings—n/l and tr/ch confusions (*Nội Bài* → *lộ tài*), *độ* (degree) → *đội*—which the model mostly handles by refusing or asking rather than acting.

### 7.2 Noise robustness

All recordings were made in a quiet room, and the system has no denoising stage. With the off-the-shelf recognizer, we mixed the 60 C+D recordings with three noises at 20, 10, 5 and 0 dB SNR relative to the speech portion: real in-car noise (DEMAND `TCAR`, Thiemann et al., 2013), synthetic brown noise and Vietnamese background speech from FLEURS (Conneau et al., 2023). The pre-registered rule was to try a denoiser only if in-car noise at 10 dB cost more than 6 of 60 commands.

In-car noise at 10 dB cost 3 commands, so no denoiser was added (Figure 4). PhoWhisper's WER

stayed at about 22% for in-car noise down to 0 dB; differences of one to three commands are within noise for 60 items. The real weakness is *another person speaking*: at 10 dB the chain lost 11 commands, and at 5 dB the recognizer transcribed the background speaker as well (WER above 100%), leaving 1 of 35 clear commands. The failure is safe: out-of-scope items stayed at 10–11 of 12, because the model refuses garbled text instead of acting on it. Single-channel denoisers target stationary noise and cannot separate a second voice; multi-microphone hardware or source separation would be needed. Music was not tested. Separately, browser audio processing (noise suppression and automatic gain) removed high-frequency consonants in the demo (*"nóng quá"* (*so hot*) heard as *lóng quá*); requesting raw audio made the demo match the measured condition.

### 7.3 Improving recognition

In live use the recognizer felt worse than the measured 63%, so we looked for the cause and for a remedy, reusing the recordings. Every remedy was selected on set B and checked on other data under a rule written down beforehand; sets C and D were reused three times in this subsection and are therefore not used for the final numbers.

**The demo's audio path is not the cause.** We replayed the 60 C+D recordings through the real interface: Chromium used each file as a fake microphone, the page recorded it with `MediaRecorder` (WebM/Opus) and posted it to the same service. The chain was correct on 36 of 60 commands, against 38 when the WAV files were sent directly (exact McNemar $p = 0.63$); half of the transcripts differed, but in both directions. The recognizer itself is the bottleneck: it misrecognizes 22% of the words even on the original files, and tiny changes in the audio change its output.

**Three training-free remedies did not help.** (i) A different recognizer: Zipformer-30M (Yao et al., 2023; hynt, 2025), a 30M-parameter transducer run with sherpa-onnx, was ten times faster (median 0.17 s vs. 1.75 s) and had a lower WER (15.7% vs. 22.0% on C+D), yet produced *fewer* correct commands (20 vs. 22 on B; 32 vs. 38 on C+D, $p = 0.18$). Its errors fall on the words that select a function—*nắp xăng* (fuel flap) → *lắp sang*, *cốp* (trunk) → *cúp*—whereas PhoWhisper's fall on sentence-final particles and filler words that the language model ignores. It failed the pre-registered rule (three more correct commands on B); a rule relaxed after seeing set B (lower WER, not significantly worse) would have admitted it, but with fewer correct commands on both sets we kept PhoWhisper. As with PhoWhisper-medium above, average WER is the wrong yardstick for spoken commands. (ii) Running both recognizers and keeping the transcript on which the language model is more confident gave exactly PhoWhisper's score (22 of 40; 38 of 60), although an oracle choice would reach 25 and 42: the decision-token confidence tells whether a sentence is a clear command, not whether it was heard correctly, so the model is just as confident on *lắp sang*. (iii) Repairing misheard multi-syllable part names with the initial-consonant confusions common across Vietnamese regions (l/n, s/x, ch/tr, d/gi/r), from a vocabulary written before any recording, changed three PhoWhisper transcripts and no outcome (on 30,060 accented training sentences it altered 0.37%, mostly injected typos).

**Fine-tuning the recognizer.** We then fine-tuned PhoWhisper-small on in-domain audio: 5,000 command sentences from the language model's *training* data (contamination-checked against all test sets and recording sets), with digits spelled out and read by 23 of the 25 VieNeu voices; 1,500 FLEURS training utterances to preserve general Vietnamese; and, for variant B only, 150 new sentences read by the first author (12 minutes, repeated four times). Each sample is degraded at random while training: speed 0.9–1.1, synthetic reverberation, DEMAND in-car noise at 0–20 dB SNR, a laptop-microphone band-pass and gain. Training takes three epochs at learning rate $10^{-5}$ (6–8 minutes on an A100); each epoch is converted to the same CTranslate2 int8 format, so latency is unchanged. The checkpoint was chosen on set B (ties by lower WER) and evaluated once on a new 40-sentence set E recorded for this purpose; forgetting was checked on 300 FLEURS test utterances and on set G, 60 frozen-test commands read by the two VieNeu voices held out from training.

Table 13 shows that synthetic speech alone barely helps on a real voice (and raised wrong actions on set E from 6 to 9), while adding 12 minutes of real recordings cuts the word errors by 40% and recovers half of the gap to typed input (24 → 28 of 32), with wrong actions unchanged at 6. What the off-the-shelf recognizer lacked was therefore

| Recognizer | Set B /40 | Set E (final) /40 | Set E (final) WER | FLEURS WER |
|---|---|---|---|---|
| Typed (same chain) | 30 | 32 | – | – |
| PhoWhisper-small | 22 | 24 | 24.8 | 11.6 |
| + TTS and FLEURS (A) | 23 | 25 | 21.2 | 10.7 |
| + 12 min real voice (B) | **25** | **28** | **14.5** | 10.8 |

Table 13: Fine-tuning the recognizer: end-to-end correct commands with the deployed language model, and word error rate (%). Set B selected the checkpoint; set E was used once.

acoustic—this voice and this microphone—rather than vocabulary. Variant B is deployed. Three cautions apply. The gain on set E is five commands won and one lost ($p = 0.22$), not significant with 40 items, though consistent with set B. Set E shares its speaker, microphone and recording session with the fine-tuning recordings, so it measures the benefit for *this* speaker. And for other speakers we could only show absence of harm: FLEURS WER did not rise, and on set G correct commands went from 29 to 46 of 60 for both variants—a figure inflated by the shared TTS engine, which we read only as "no harm". A check on other real speakers giving commands is missing; the noise and wake-word experiments were run with the original recognizer.

**Invented text on background noise.** The fine-tuning data contain no audio without speech, and the VAD gain described in §7.4 occasionally lets background noise through as speech (8 of 190 ten-second background segments cut from our recordings, against 0 without the gain). On such segments the fine-tuned recognizer invented a whole sentence every time (70 of 70). Whisper's no-speech probability was always zero, but the average token log-probability separates the two cases: $-0.99$ to $-0.57$ on background noise, $-0.22$ to 0 on real commands. Transcripts below $-0.4$ are discarded as "not heard". On all 140 real commands of sets B–E none is discarded (lowest $-0.26$), so every number above is unchanged. The threshold was set on background noise from the same recordings; a television or a passenger talking is not filtered.

### 7.4 Wake word

"Hey Nexa" lets the driver start a command hands-free. Porcupine's free tier is no longer available and running STT continuously costs about 1.8 s of CPU per utterance, so we trained a detector on

| | v1 | v2 | Target |
|---|---|---|---|
| Real calls detected (40, new set) | 16[a] | **39** | ≥36 |
| False alarms / h (FLEURS test, 3.15 h) | **1.27** | 2.86 | ≤1 |
| Triggers on 30 spoken car commands | – | 1 | 0 |
| Self-wake on assistant replies (38) | 0 | 0 | 0 |

Table 14: Wake-word test (each measured once). [a]On the first recording set. Rules: v1 four frames $\geq 0.05$; v2 three frames $\geq 0.85$. CPU: about 3.5% of one core.

openWakeWord's (Scripka, 2023) frozen feature extractors (mel spectrogram and a speech embedding model, Lin et al., 2020): 16 frames of 96 dimensions (1.28 s) feed a small scikit-learn MLP, updated every 80 ms. A detection requires $k$ consecutive frames above a threshold, followed by a 2 s refractory period; Silero VAD (Silero Team, 2024) ends the command after silence, and the wake phrase is stripped from the transcript. Wake-word commands carry the source `wake_word`, which activates the confirmation rules of Table 2; the detector pauses while the assistant speaks.

Version 1 used only synthetic positives (VieNeu reading six Vietnamese spellings of "hey nexa" with 20 voices, 5 held out), synthetic negatives (near-miss phrases, car commands, the assistant's own replies), Vietnamese FLEURS training audio and 500k precomputed ACAV100M feature windows (Lee et al., 2021). It caught 98.9% of held-out synthetic calls but only 16 of 40 of the first author's real calls. The recordings peaked at $-23$ to $-38$ dBFS, whereas training positives peaked at $-18$ to $-3$ dBFS, and fast calls rarely stayed above threshold for the required four frames. Version 2 widened the level augmentation to $-46 \ldots -3$ dB, added 46 real training calls (14 held out for validation, none from the test recordings), and re-selected $k$ and the threshold on validation.

Version 2 detects almost every real call in every style (normal, fast, quiet, from 1–1.5 m, with *ơi*, followed directly by a command), but false alarms doubled on continuous read speech, and one of the author's car commands triggered it (Table 14). We deploy version 2 and report that it meets the detection target but not the false-alarm target. If nobody speaks for 2 s after a detection, the call is treated as a bare "Hey Nexa": the assistant greets the driver and listens for 10 s, then returns to waiting for the wake word. A false wake-up therefore produces an audible greeting, or sends the next overheard sentence to the assistant, which usually refuses it; sensitive functions always require confirmation for

wake-word commands. After every answer the service also listens for 10 s for a follow-up without the wake word.

**Quiet microphone.** In live use, commands spoken after the wake word were sometimes ignored or cut off. The laptop microphone records quietly (peaks around −27 dBFS) and Silero VAD missed speech: replaying the 190 real command recordings (set E and the fine-tuning set) through the command capture, only 40 were captured whole, 103 were cut mid-sentence and 47 were dropped. The service now amplifies the signal *for the VAD only*, with an adaptive gain of 1–8× that brings the recent peak to −12 dBFS (20 s half-life); the recognizer still receives the original audio, so no STT number changes. On the same recordings 164 commands were captured whole, 26 were cut and none was dropped. The gain was chosen on these recordings (one speaker, one microphone). Because ACAV100M features are licensed CC BY-NC-SA, the detector may only be used non-commercially.

### 7.5 Speech synthesis

TTS is used off the shelf because it does not affect what the car does: it only reads the template answer after the decision has been made. We still checked its intelligibility automatically. Every utterance of the four single-turn test sets was run through the runtime with the reference call as the model output, which gives the 163 distinct answers the assistant actually speaks (80 of them contain numbers such as *260,4 km* or *62%*). VieNeu read them with the deployed voice and two recognizers transcribed them: the off-the-shelf PhoWhisper-small, which has never heard VieNeu (the fine-tuned one was trained on its voices), and Zipformer-30M as a second judge. Both sides are normalized to spoken form (digits to words, one tone-mark convention, no punctuation). On the 143 answers that contain only Vietnamese syllables, the word error rates are 3.1% and 1.2% (74% and 88% of the answers transcribed exactly), lower than PhoWhisper's 11.6% on human read speech (FLEURS). On the 20 answers with foreign words, names or identifiers they are 26.9% and 18.6%: the assistant's name *NexaAI*, *Playlist*, *VOV* or *PSI* are read with Vietnamese spelling rules. The check also exposed a template error: nearby-search answers read out the simulator's internal identifiers (*charging_station_1 — 3.0 km*). Intelligibility measured by a recognizer is not naturalness; we ran no listening test.

### 7.6 Latency

On the laptop CPU a spoken command takes about 2.5 s from end of speech to the start of the spoken answer: STT about 1.8 s for a 2–5 s utterance, the model about 0.5 s, and TTS about 0.25 s until the first audio. The answer is synthesized and streamed chunk by chunk, and the interface plays each chunk while the next one is generated. On ten answers this cut the time to first audio from a median of 798 ms (worst 2.8 s for a three-part answer) when waiting for the whole sentence to 241 ms (worst 285 ms). Synthesis runs faster than playback (0.85 s for 2.7 s of audio), so chunks play without gaps; a later measurement on 20 answers gave 207 ms to first audio (worst 249 ms) and a real-time factor of 0.37–0.41. Each extra segment of a multi-command utterance adds about 0.5 s. The laptop's 4 GB GPU is not used. These numbers require the laptop's performance power mode: in its default mode the CPU ran at about 60% of its nominal clock during synthesis, the real-time factor rose to 1.1 and the first audio to about 0.55 s, so speech was produced slower than it was played and playback stuttered.

## 8 Discussion

**Where the small model stands.** On the commands a car assistant mostly receives, the 1.5B student is statistically indistinguishable from the 72B teacher while being 65 times cheaper in prompt length, about eight times faster on the same GPU and small enough for a laptop CPU. Its consistent weakness is knowing what it *cannot* do: unsupported features that resemble supported ones ("honk", "lane-departure warning", "left turn signal" → hazard lights at $p = 0.99$) are mapped to the nearest real function, often with high confidence. Its second weakness is context: given the same previous call, the prompted teacher resolves two-turn follow-ups clearly better (81.7% vs. 62.5%), so the 4.7k training examples with a previous call (about 1.5k of them genuine follow-ups) taught the format more than the reasoning. Data helps (contrastive refusals, more samples) and a threshold helps against uncertain mistakes, but neither removes confidently wrong decisions; this matches the over-triggering reported for functional tokens by Jafarzadeh Asl et al. (2026). A hybrid design that forwards uncertain requests to a larger model when a connection is available is the natural next step for a product.

**Form sensitivity is a distillation artifact.** The student reacts to surface form that carries no meaning: lower-casing only the first letter of the 230 clean-test sentences raised false refusals of the v5 model from 3 to 21, and a lower-case *"hỏng xe giữa đường rồi"* (*my car broke down*) was refused while the capitalized version was understood. The teacher writes clean, capitalized, formal sentences, and sequence-level distillation transfers only what appears in the data. The same cause explains the gap between validation (97–99%) and the independently generated tests (75–88%), the lexical lock-in of R1, and the need for short and noisy data in v6. Diversity has to be designed into generation (assigned personas, contexts, lengths and vocabulary), because asking the teacher to "write many varied sentences" does not produce it.

**Shortcuts appear wherever a feature is one-sided.** Three times a feature of sentence form ended up on one side of the label distribution—imperative mood on calls (R1), stories on refusals (R4), long sentences on refusals rather than clarifications (before v5)—and each time the model learned it as a rule. Adding hard examples in contrastive pairs across labels was the single most effective data intervention we found. A test set of mostly short commands did not reveal the R4 shortcut; only a development set containing story-like *doable* requests did.

**Measurement discipline changed conclusions.** Several results looked like improvements and were not: R2's gain on the ambiguous group was one lucky seed; the "student beats 72B on ambiguity" impression came from a 39-item group and reversed on 130 items; the threshold's cost on clear commands was 6–10 times larger on test than on validation; 4-bit quantization passed on validation and lost five points on implicit requests at test time. Freezing test sets before scoring, training two seeds, testing paired outputs and writing criteria down in advance was cheap compared with the cost of drawing these wrong conclusions.

**When RL helps.** GRPO needs prompts on which the policy sometimes succeeds and sometimes fails, and benefits from a reasoning step that offers several paths to sample. A functional-token model trained by distillation offers neither: its decision is one token, and most training prompts are already solved. RL on fresh, harder prompts or a preference method on explicitly paired outputs (Ross et al., 2025) would be better matched to this setting; we did not run either.

## 9 Conclusion

Nexa Drive shows that a 1.5B model with functional tokens, distilled from an open 72B teacher with about US$100 of rented GPU time, can serve as a Vietnamese in-car assistant on a laptop CPU: it handles clear and implicit requests about as well as the prompted teacher, asks instead of guessing when a request is only a complaint, and is wrapped in code that keeps wrong decisions away from the car. The project's main lessons are about data and measurement rather than architecture: hard examples must be added in contrastive pairs, synthetic validation data hides the weaknesses that matter, and small test sets need several seeds and paired tests before a difference is believed. The remaining weaknesses—refusing unsupported features, two-turn follow-ups, sensitivity to quantization on ambiguous input, background speech, and false wake-ups—are measured and documented, and define the next steps: real-user data, a hybrid on-device/cloud design and multi-microphone input.

## Limitations

**Synthetic evaluation.** All language test sets were generated by GPT6-Astra from a written specification and reviewed by one person, the first author, who also saw the draft labels. They are separate from the training generator but are still LLM text: real drivers speak more tersely, hesitate and make requests no specification anticipates. Absolute accuracies may therefore be optimistic; comparisons between systems scored on the same sets are more reliable. The two reviewed label sets used to measure the judges (GPT6-Astra and Claude drafts) were corrected by the same person, so we have no independent human–human agreement baseline.

**Small sets and repeated use.** Groups contain 28–150 items and the clean test was used across many rounds; v6 and v6.1 were two attempts on the same test, and the Ambig-3 set was retired after one use. We trained only two seeds per configuration (three for the final one; seed standard deviations reach 5–6 points on the smaller groups, Table 18). Differences below about five points should be read as unresolved.

**One speaker, quiet room.** All real recordings (STT sets, wake-word training and test calls) come from the first author on a laptop microphone. Noise was added digitally from recordings, not captured while driving; music was not tested. The wake word was trained partly on the author's voice and may detect other speakers less reliably; it is not speaker verification. The same holds for the fine-tuned recognizer: its gain was measured on the author's voice, and for other speakers we only verified that general Vietnamese (FLEURS) and held-out synthetic voices did not get worse; we could not obtain a multi-speaker set of real spoken commands in time. The fine-tuned recognizer never saw audio without speech; a confidence filter stops it from inventing text on background noise, but not on a passenger or a radio talking. The microphone gain for the VAD was tuned on the same speaker and microphone.

**Scope.** The vehicle is simulated; navigation, nearby search and weather return simulated results. The system is Vietnamese only (an English extension was planned but not done). The model cannot execute several independent actions in one call—multi-command utterances are split by connector words at runtime, and requests joined by *và* ("and") still lead to a single action. Answers are template-based by default; the speech synthesizer reads foreign words and names with Vietnamese spelling rules. Closing the fuel flap or charge port is handled by a hand-written rule, not by the model.

**Thresholds and quantization.** The threshold and the compression level were chosen on data whose distribution differs from the tests; on test, the threshold cost more clear commands and quantization more ambiguous items than validation predicted. No model could be certified to keep wrong behavior below 10% on the calibration set.

**Guardrails.** The safety rules are tested by scenario tests but not measured on a benchmark; they cover a handful of state conditions and do not include cross-component rules (e.g. AC and windows), weather or emergency hand-over.

## Ethics Statement

The assistant cannot steer, brake or accelerate; actions that open the car are refused while driving and confirmed when requested through the wake word. Voice recordings are of the first author only, who consented to their use; the audio is not publicly released, only transcripts and scores are. External data are used under their licenses: FLEURS (CC BY 4.0), DEMAND (CC BY-SA 3.0), ACAV100M features (CC BY-NC-SA 4.0, which restricts the wake-word model to non-commercial use), the Khronos *Car Concept* 3D model (CC BY 4.0). VieNeu-TTS embeds an inaudible watermark in synthetic speech. GPT6-Astra was used to generate evaluation sentences and draft labels, which were reviewed by a person. An AI assistant (Claude) was used for code assistance and to draft this report from the project log; all reported numbers were checked against the experiment output files.

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## A Function Inventory

Table 15 lists the 24 functions with their arguments (`?` optional). Read-only functions (R) are never clarified; sensitive functions (S) are never proposed from an implicit request and require confirmation from the wake word.

## B Prompts and Outputs

**Call prompt (turn 1).** The system message reads *Bạn là tác tử điều khiển xe ô tô.* ("You are an agent that controls a car.").

```
<|im_start|>system
Bạn là tác tử điều khiển xe ô tô.<|im_end|>
<|im_start|>user
Bật điều hòa 22 độ ghế lái<|im_end|>
<|im_start|>assistant
```

Training target (in most examples it ends here; 7.1% additionally continue with a function description after `<nexa_end>`, which is never used at inference):

```
<nexa_6>(state="on", temperature=22,
  zone="driver")<nexa_end>
```

Optional continuation: `Mô tả hàm: Bật hoặc tắt điều hoà, …` ("function description: turn the AC on or off…").

**Two-turn prompt (v6 on).** The user turn starts with the last executed call; read-only calls are not carried over.

```
Lượt trước: <nexa_15>(window="front_right",
  position="down", amount=100)
Bên trái phía trước cũng vậy luôn nha.
```

("Previous turn: lower front-right window. Do the same for the front left.") →

```
<nexa_15>(window="front_left",
  position="down", amount=100)
```

**Clarification and refusal.** *"Trong xe bí quá"* (*it is stuffy in the car*) → two proposals separated by a semicolon, rendered as "Turn on the AC or lower the windows?":

```
<nexa_ask><nexa_6>(state="on");
<nexa_15>(window="all",
  position="down")<nexa_end>
```

*"Bạn có thể giúp tôi sửa xe được không"* (*can you fix my car*) → `<nexa_24>()<nexa_end>`.

**Answer prompt (turn 2).** The call is followed by a `tool` turn holding the JSON result; the target is a Vietnamese sentence written by the teacher that may only use numbers present in the result.

**Schema-in-prompt baseline.** The baseline prompt (2,156 tokens) contains the descriptions and arguments of the 23 functions, the three allowed JSON actions (`call`, `clarify`, `refuse`), the same data policy as the training data (no clarification for read-only functions, no proposal of sensitive functions, only stated arguments) and three examples that do not occur in any test set. Outputs are converted to the functional format; invalid JSON or unknown functions are scored as unparseable.

## C Data Rounds in Detail

Table 16 lists every configuration. Further details:

- **Contamination**: after R1, 43 of the original 296 test items overlapped with generated data (66 after later rounds and seed examples were included); on the overlapping items R1 scored 18 points higher on clear commands than on the rest. Only the clean 230 items are reported. Two of our own seed examples in the schema had been written from memory of test items; an automated test now blocks any overlap between seed examples and test sets.

| Token | Function | Arguments | |
|---|---|---|---|
| 0 | `get_fuel_level` | – | R |
| 1 | `get_battery_range` | – | R |
| 2 | `get_tire_pressure` | `wheel?` ∈ {front_left, front_right, rear_left, rear_right} | R |
| 3 | `get_vehicle_status` | – | R |
| 4 | `get_odometer` | `unit?` ∈ {km, mi} | R |
| 5 | `get_range_estimate` | `trip_km?` $\geq 0$ | R |
| 6 | `set_ac` | `state` ∈ {on, off}, `temperature?` 16–30, `zone?` ∈ {driver, passenger, rear, all}, `delta?` $-14..14$, `mode?` ∈ {cool, heat, auto} | |
| 7 | `set_fan_speed` | `level?` 0–7 or `delta?` $-7..7$ (exactly one) | |
| 8 | `set_defrost` | `state` ∈ {on, off}, `zone?` ∈ {front, rear, both} | |
| 9 | `lock_doors` | – | |
| 10 | `unlock_doors` | `door?` ∈ {driver, passenger, rear_left, rear_right, all} | S |
| 11 | `open_trunk` | – | S |
| 12 | `close_trunk` | – | |
| 13 | `open_charge_port` | – | S |
| 14 | `open_fuel_flap` | – | S |
| 15 | `control_window` | `window` ∈ {front_left, front_right, rear_left, rear_right, front, rear, all}, `position` ∈ {up, down}, `amount?` 0–100 (default 100) | |
| 16 | `operate_sunroof` | `action` ∈ {open, close, tilt}, `intensity?` 0–100 | |
| 17 | `set_headlights` | `mode` ∈ {off, auto, low_beam, high_beam, parking} | |
| 18 | `set_hazard_lights` | `state` ∈ {on, off} | |
| 19 | `set_volume` | `set_value?` 0–40 or `delta?` $-40..40$ (exactly one) | |
| 20 | `play_media` | `source?` ∈ {radio, bluetooth, usb, streaming}, `query?` | |
| 21 | `navigate_to` | `destination`, `avoid?` ∈ {none, tolls, highways} | |
| 22 | `find_nearby` | `category` ∈ {gas_station, charging_station, parking, restaurant, hospital, atm, coffee}, `radius_km?` 0.5–25 | |
| 23 | `get_weather` | `location?`, `day?` 0–6 | R |
| 24 | `irrelevant_function` | – (refuse) | |

Table 15: Function signatures of the final schema (v6). `delta`/`mode` of `set_ac`, `delta` of `set_fan_speed` and the `front`/`rear` windows were added in round v6, appended at the end of the argument list so that older positional calls keep their meaning.

- **Judge failures we observed**: Llama demanding optional arguments; Qwen rejecting hard refusals that sound close to real functions ("recirculate air", "open the front trunk"), because the judging prompt listed only function *groups*—fixed by giving exact signatures and a calibration gate (hard refusals kept 22/28 → 28/28); Qwen accepting nearly every clarification.
- **Teacher drift during generation**: asked for complaints about the fan, the teacher wrote complaints about heat (only 14% of fan sentences kept their target function in R2). Situation guidance per function ("what to write" and "what belongs to another function", without example sentences) raised this to 83% in the next generation.
- **Near-duplicate filtering at scale**: a word-overlap threshold chosen on 64 sentences discarded 2,962 of 4,700 v5 sentences when applied across all functions; re-applying it within each function only (threshold 0.45) recovered 1,665.
- **Conventions decided during labeling**: a complaint about cold *with* the AC mentioned means turning it off, without it means heating; "volume to $N$" sets the value, "volume up by $N$" changes it; strong or noisy airflow means fan level 2 rather than off. Each convention was applied to training data by code and to test labels before the corresponding test was frozen.

## D Per-Seed Results

Table 17 gives the per-seed comparisons behind §6.1 and §6.4. With the threshold 0.85, the v6.1 vs. v5 comparison on out-of-scope was 85.5 vs. 88.9 ($p = 0.29$) and 78.6 vs. 87.2 ($p = 0.013$), so the threshold did not close the gap. A risk–coverage analysis over all thresholds on the clean test showed the same: at $\leq$6% over-triggering, v5 executes 70.9% of clear commands immediately and v6.1 only 18.2% (mean of seeds), whereas over *all* test sets, including short, two-turn and noisy commands, the area under the risk–coverage curve favors v6.1 (19.6 vs. 24.7, lower is better).

Table 18 adds a third seed. The two-seed means move by at most 3 points (ambiguous 75.6 → 72.6, typos 63.5 → 60.6), and seed-to-seed standard deviations reach 5–6 points on the 39-item ambiguous group, out-of-scope requests and typos, confirming that single-seed differences of this size are

| Config. | Train | Change (diagnosis from training statistics or validation) | Clear | Amb. | OOS |
|---|---|---|---|---|---|
| R1 | 15,942 | pool of 27k teacher samples, 300 per function, 1:1 refusals | 70.3[1] | 59.0[1] | 63.2[1] |
| R2 | 16,080 | synonyms (`tu_dong_nghia`) for 15 functions, 1,667 imperative refusals, jury-filtered clarifications, situation guidance for 6 thin functions | 75.0 (4.0) | 65.4 (12.8) | 61.1 (7.7) |
| R3 | 16,010 | code fixes: fuel range, volume target, window direction, duplicate labels | 76.3 (1.3) | 66.7 (5.1) | 57.2 (5.1) |
| +desc | 16,010 | function description appended to targets (Octopus v2) | 79.7 (0.0) | 70.5 (2.6) | 68.0 (11.1) |
| +1k | 28,341 | up to 1,000 samples per function | 75.7 (2.7) | 65.4 (2.6) | 69.2 (1.7) |
| R4-repl. | 15,994 | imperative refusals replaced by 1,651 story-like refusals (rejected) | 75.7 (2.7) | 65.4 (2.6) | 58.6 (19.7) |
| R4-add | 17,661 | story-like refusals added (rejected) | 75.7 (0.0) | 73.1 (2.6) | 59.4 (0.8) |
| Contrastive | 17,635 | R4-repl. + 1,641 story-like calls | 78.4 (5.4) | 69.2 (10.3) | 74.8 (7.7) |
| Combined | 29,966 | contrastive + 1k/function + descriptions | 78.3 (2.7) | 62.8 (2.6) | 73.0 (0.9) |
| v5 (full) | 19,322 | contrastive + 1,696 story-like clarifications; fan convention fixed | 79.1 (1.3) | 76.9 | 75.6 (6.0) |
| v5 (LoRA) | 19,322 | same data, LoRA | 79.1 (1.3) | 76.9 | 83.3 (2.5) |
| v6.1 (LoRA) | 33,523 | see §6.4 | 78.4 (5.4) | 75.6 (2.5) | 77.4 (11.1) |

Table 16: All configurations trained, clean test, mean of two seeds with the spread between seeds in parentheses ([1]one seed). Full fine-tuning unless marked. R4 variants were rejected because they refused 4–8 of 10 doable story-like development requests.

| | Seed 42 | | Seed 7 | |
|---|---|---|---|---|
| v6.1 vs. 72B | v6.1 / 72B | $p$ | v6.1 / 72B | $p$ |
| Clear (74) | 81.1 / 83.8 | 0.77 | 75.7 / 83.8 | 0.15 |
| Ambiguous (39) | 74.4 / 56.4 | 0.065 | 76.9 / 56.4 | 0.039 |
| Out-of-scope (117) | 82.9 / 94.9 | 0.001 | 71.8 / 94.9 | <0.001 |
| Supplement (28) | 57.1 / 39.3 | 0.27 | 60.7 / 39.3 | 0.18 |
| Ambig-4 (129) | 79.8 / 70.5 | 0.065 | 78.3 / 70.5 | 0.14 |
| Short + paraphr. (120) | 88.3 / 91.7 | 0.39 | 87.5 / 91.7 | 0.27 |
| New arguments (30) | 83.3 / 70.0 | 0.34 | 83.3 / 70.0 | 0.34 |
| Two-turn (60) | 66.7 / 81.7 | 0.049 | 58.3 / 81.7 | 0.001 |
| Typos (104) | 60.6 / 69.2 | 0.093 | 66.3 / 69.2 | 0.66 |
| v6.1 vs. v5 | v6.1 / v5 | $p$ | v6.1 / v5 | $p$ |
| Short + paraphr. (120) | 88.3 / 75.8 | 0.0007 | 87.5 / 76.7 | 0.007 |
| Two-turn (60)[a] | 66.7 / 23.3 | <0.001 | 58.3 / 23.3 | <0.001 |
| Clear (74) | 81.1 / 78.4 | 0.69 | 75.7 / 78.4 | 0.69 |
| Out-of-scope (117) | 82.9 / 84.6 | 0.73 | 71.8 / 82.9 | 0.007 |
| Ambig-4 (129) | 79.8 / 78.3 | 0.79 | 78.3 / 78.3 | 1.00 |
| Typos (104) | 60.6 / 31.7 | <0.001 | 66.3 / 31.7 | <0.001 |

Table 17: Exact McNemar tests per seed (unquantized, no threshold). [a]v5 with rule-based memory.

| v6.1, unquantized | s42 | s7 | s2026 | Mean | SD |
|---|---|---|---|---|---|
| Clear (74) | 81.1 | 75.7 | 79.7 | 78.8 | 2.8 |
| Ambiguous (39) | 74.4 | 76.9 | 66.7 | 72.6 | 5.3 |
| Out-of-scope (117) | 82.9 | 71.8 | 78.6 | 77.8 | 5.6 |
| Supplement (28) | 57.1 | 60.7 | 57.1 | 58.3 | 2.1 |
| Ambig-4 (129) | 79.8 | 78.3 | 76.7 | 78.3 | 1.6 |
| Short + paraphr. (120) | 88.3 | 87.5 | 88.3 | 88.1 | 0.5 |
| New arguments (30) | 83.3 | 83.3 | 76.7 | 81.1 | 3.8 |
| Two-turn (60) | 66.7 | 58.3 | 65.0 | 63.3 | 4.4 |
| Typos (104) | 60.6 | 66.3 | 54.8 | 60.6 | 5.8 |
| Wrong actions (clean test) | 37 | 51 | 43 | 43.7 | 7.0 |

Table 18: A third seed (2026) for the final configuration, added after the main analysis only to estimate seed variance; identical data and hyper-parameters, checkpoint chosen by the same validation rule (it selected epoch 1, ck2096, where all three epochs tied on clear commands). No decision was revised.

noise. The deployed seed 42 has the fewest wrong actions and the best out-of-scope accuracy of the three, so deployed-system numbers are somewhat optimistic relative to an arbitrary seed; it was selected on the calibration set, not on these scores.

## E Runtime Details

- **Normalization** (`serve/chuan_hoa.py`): collapse whitespace; capitalize the first letter; collapse a repeated final consonant (*cốpp* → *cốp*; Vietnamese has no double final consonants) and letters repeated three or more times; lower-case sentences that are at least 80% capitals. Digits are untouched.
- **Speech normalization**: spoken numbers to digits, including colloquial two-digit forms (*hai hai* = 22, *hai lăm* = 25) next to a unit; *không* ("zero"/"not") becomes 0 only after "level" or "number"; hundreds/thousands without a preceding digit are kept as words when not followed by a unit (*bà Triệu* is a street).
- **Relative conventions**: temperature ±2 °C, fan ±1, volume ±3 when no number is given; "high beam" only when stated; a number directly before "window(s)" is a count, not a percentage. When a relative command does not change the value because a limit is reached, the answer says so ("the fan is already off").
- **Multi-command connectors**: *xong, xong đó, xong rồi, xong thì, sau đó, tiếp theo, tiếp đó, với cả, rồi thì*, and *, rồi*; each segment must have at

| Item | Value |
|---|---|
| Student | Qwen2.5-1.5B-Instruct |
| Teacher / judges | Qwen2.5-72B-Instruct-AWQ, Llama-3.3-70B-AWQ |
| Optimizer | AdamW, lr $5 \times 10^{-5}$, 10 warm-up, linear |
| Epochs / batch / max len | 3 / 16 (effective) / 256 |
| LoRA | r 16, $\alpha$ 32, q,k,v,o,up,down + full emb./head |
| Seeds | 42, 7 |
| GRPO | $G$ 8, lr $10^{-5}$, $\beta_{\text{KL}}$ 0.04, $T$ 1.0, 2 epochs |
| Quantization | llama.cpp q4_K_M (output layer Q6_K) |
| Inference | llama-server, CPU 8 threads, greedy, 64 tokens |
| Hardware (train) | A100 40/80 GB, L40S, L4 (Modal) |
| Hardware (run) | laptop, Intel i5-12500H, CPU only |
| Total GPU cost | about US$100 |

Table 19: Hyperparameters and compute.

least two words; segments are processed in order, calls and state updates are merged into one answer, and processing stops at the first clarification, which also lists the remaining segments.

- **Social replies**: short greetings, thanks, farewells, compliments and questions about the assistant's name or abilities ($\leq$5 words, no control verb) get a fixed friendly reply without calling the model; other refusals get a neutral apology.
- **Pending call**: after a clarification, "yes"-type answers execute the stored call, "no" cancels it, and any other sentence cancels it and is processed as a new request; with two proposals the driver must say "the first" or "the second".

## F Hyperparameters and Compute

Table 19 summarizes the configuration. Two lessons from the infrastructure: files downloaded from the cloud volume were silently corrupted or truncated several times even when the client reported success, so every model file is checked against a SHA-256 computed on the server before use; and the base model's tied embeddings had to be untied before LoRA training (§5.1), which we verified with a tiny model before the first real run.

## G Speech Details

Table 20 gives all speech results. For the noise experiment, each noise segment is drawn at a random offset with a fixed seed and scaled so that the ratio between the power of the speech portion (silence trimmed at 35 dB) and the noise matches the target SNR. STT latency grows slightly with noise (median 1.7 $\rightarrow$ 2.0 s).

| Set | Typed | Parakeet | Ph-small | Ph-medium | Type |
|---|---|---|---|---|---|
| A (230, TTS) | 185 | 148 | 151 | 155 | synthetic |
| B (40) | 30 | 15 | 21 | 18 | selection |
| B, after fix | 30 | 16 | 22 | 19 | post-hoc |
| C (30) | 24 | 14 | 21 | 19 | new |
| D (30) | 18 | 12 | 17 | 15 | new |

Table 20: End-to-end correct commands per recognizer and set (v6.1 seed 42, q4, threshold 0.80; set A row after the normalization fix). WER of PhoWhisper-small: A 17.9%, B 20.2%, C 23.0%, D 20.9%.

The wake-word detector uses the streaming mel-spectrogram and embedding ONNX models of openWakeWord (about 3 ms per 80 ms frame) and a scikit-learn MLP trained with `partial_fit` on batches mixing our negatives (30%), ACAV100M features (35%) and Vietnamese FLEURS training features (35%). Validation contained 27.5 minutes of held-out negative speech; $k$ and the threshold were chosen as the pair with at most one false alarm per hour on it and the highest detection on held-out real calls. The first selection rule (a single frame above threshold) could not reach one false alarm per hour on validation at any threshold; the $k$-consecutive rule was introduced after seeing validation results but before any test measurement. FLEURS test was used for both versions, so a new negative test set would be needed for further versions.

## H Error Examples

- **Over-triggering**: *"Bật xi nhan trái"* (*turn on the left turn signal*) $\rightarrow$ hazard lights ($p = 0.99$); *"Bật khoá vi sai"* (*engage the differential lock*) $\rightarrow$ unlock the passenger door; *"Hạ riêng tiếng bass xuống hai nấc"* (*lower only the bass by two steps*) $\rightarrow$ volume $-2$.
- **Clarification on chit-chat**: *"Nhìn hàng phượng này lại nhớ hồi đi học ghê"* (*seeing these flame trees reminds me of school*) $\rightarrow$ "open the sunroof?" ($p = 0.96$).
- **Speech**: *"vặn nhỏ loa"* (*turn the speaker down*) heard as *vận nhỏ là*; *"Hủy"* (*cancel*) heard as *Quỳ*, which led to an over-trigger; "26 degrees" heard as *hay xấu thủ*.
- **Fixed by data**: *"Hạ kính ghế phụ"* (*lower the passenger window*) $\rightarrow$ defrost (R1); *"bật pha"* (*high beam on*) $\rightarrow$ refused (v5), correct in v6.1; *"giảm nhiệt độ điều hòa"* (*lower the AC temperature*) $\rightarrow$ AC off (v5), `delta=-2` in v6.1.